

A Method of Determining the Permeability and Effective Porosity of Unconfined Anisotropic Aquifers

G. DAGAN

Technion Israel Institute of Technology, Haifa

Abstract. The equations of unsteady flow toward a partially penetrating well in an unconfined aquifer of finite thickness are solved by linearization. It is assumed that the aquifer is nondeformable and that the effective porosity at the water table is constant. It is also assumed that the aquifer is anisotropic (the principal axes of the permeability tensor being horizontal and vertical, respectively), that the pumping discharge is constant, and that the drawdowns are small. The vertical component of the flow velocity is not neglected. The solution is, therefore, equally valid in the vicinity and at large distances from the pumping well. In the latter case, it coincides with the Theis solution. A method of matching the theoretical solution with drawdown measurements during pumping tests is outlined. By the matching method the horizontal and vertical hydraulic conductivities and the effective porosity of the aquifer may be determined. The method is illustrated by analyzing data from pumping tests carried out in three anisotropic aquifers. (Key words: Drainage; ground water; hydrology; permeability; piezometers; porosity; porous mediums; seepage; water)

LIST OF SYMBOLS

D ,	initial aquifer constant depth;	z ,	depth of observation well below initial water table (negative);
$H(r', z', t')$,	auxiliary function;	$z' = z/L$,	depth of observation well below initial water table (dimensionless);
K ,	hydraulic conductivity of isotropic aquifer;	ϵ ,	small parameter;
K_h ,	horizontal hydraulic conductivity;	$\varphi = p/\gamma + z = -s$,	head;
K_v ,	vertical hydraulic conductivity;	$\xi(\rho_0, t' - \tau)$,	auxiliary function;
l ,	well's filter length;	$\phi = -K\varphi$,	velocity potential;
$l' = l/L$,	well's filter length (dimensionless);	$\phi' = \phi/K_h L = -\varphi/L$,	velocity potential (dimensionless);
l_0 ,	the depth of the well's filter center;	$\nabla = \frac{\partial}{\partial r'} + \frac{\partial}{\partial z'}$,	gradient
$l_0' = l_0/L$,	the depth of the well's filter center (dimensionless);	$\nabla^2 = \frac{1}{r'} \frac{\partial}{\partial r'} + \frac{\partial^2}{\partial r'^2} + \frac{\partial^2}{\partial z'^2}$,	Laplacian;
L ,	reference length ($L = D$ in the case of aquifer of finite thickness);	η ,	water table elevation;
n ,	effective porosity at the water table;	$\eta' = \eta/L$,	water table elevation (dimensionless);
Q ,	discharge;	τ ,	integration variable.
$Q' = Q/K_h L^2$,	discharge (dimensionless);		
r ,	distance of observation well from pumping well;		
$r' = (r/D)(K_v/K_h)^{1/2}$,	distance of observation well from pumping well (dimensionless);		
s ,	drawdown;		
$s' = sK_h D/Q$,	drawdown (dimensionless);		
t ,	time;		
$t' = tK_v/nD$,	time (dimensionless);		

INTRODUCTION

An unconfined aquifer of infinite horizontal extent and of finite thickness is considered. A partially penetrating well starts pumping at $t = 0$ (Figure 1). It is assumed that the aquifer is homogeneous and incompressible, that the effective porosity at the water table is con-

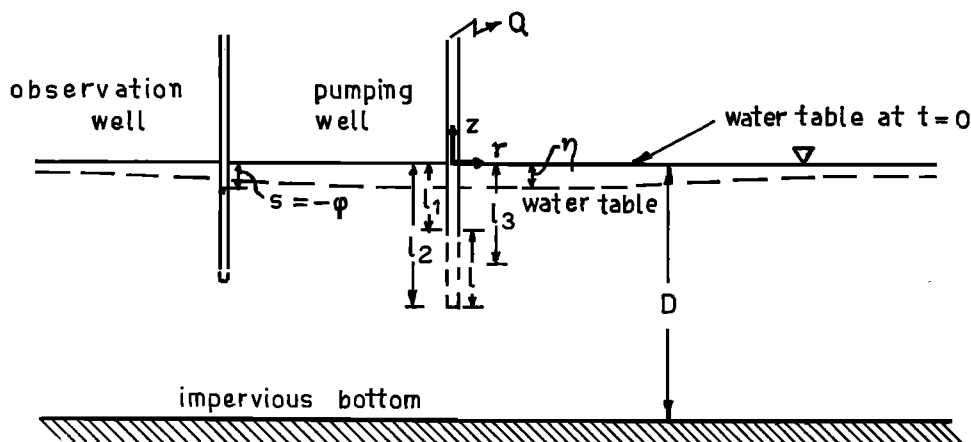


Fig. 1. A partially penetrating well in an unconfined aquifer.

stant, and that the well may be replaced by a sink line of uniform intensity.

The solution of the problem requires the determination of the head φ as a function of r , z , t and the parameters D , l_a , l , K_a , K_v , n . The solution of the equations 2-7, satisfied by φ , is very difficult, mainly because equation 3 is nonlinear and is posed on the unknown water table.

A general approximate method of solving problems of free-surface flow in porous mediums has been presented by Dagan [1964, 1966]. By this method the head is expressed as a small perturbation expansion

$$\varphi = \epsilon\varphi_1 + \epsilon^2\varphi_2 + \dots \quad (1)$$

where $\epsilon = Q/KD^2$ is chosen as a small parameter. The linearized equations satisfied by the first-order approximation φ_1 (equations 9-11) have already been used by Polubarinova-Kochina [1962] in two-dimensional flows and by Boulton [1954] in the cases of flow toward fully penetrating wells. Kirkham [1958] has obtained the same first-order equations from a physical model of the actual flow geometry. Stallman [1965] has used an RC analog to solve the problem.

The approximation of the exact solution by the first term of expansion 1 may be accurate if ϵ is sufficiently small. The computation of the second approximation φ_2 (section 2 of this paper and Dagan [1967b]) in the case of very deep aquifers ($D/l_a \rightarrow \infty$) shows that expansion 1 seems to be reasonably convergent.

The usual nonequilibrium or Theis formula

[Theis, 1935] may be obtained as a particular case of the first-order approximation φ_1 if it is assumed that the flow is essentially horizontal, i.e., $\partial\varphi_1/\partial z = 0$ [Dagan, 1966]. The Theis formula may be accurate only beyond a certain distance from the pumping well. This limiting distance depends on the well's degree of penetration and the aquifer anisotropy. Moreover, the formula may provide information only on the horizontal hydraulic conductivity of the aquifer, since the vertical velocity component is neglected.

In contrast, the linearized theory provides a solution that is equally valid in the vicinity of the well and at a large distance from the well. It may be used, therefore, to determine the horizontal and vertical hydraulic conductivities K_a and K_v of an anisotropic aquifer (it is assumed that the principal axes of the tensor of hydraulic conductivity are horizontal and vertical, respectively). Even if one is interested only in the value of K_a , the permeabilities ratio is necessary to determine the r range in which the Theis formula is valid.

The purpose of the present paper is to solve the equations of flow in the linearized approximation ($\epsilon\varphi_1$) and to provide a method of matching pumping test measurements with the theoretical results to obtain the values of K_a , K_v , and n of the investigated aquifer.

In fact, it is well known (see, for instance, Bear and Dagan [1965]) that the solution of the flow in an anisotropic aquifer is obtained by a simple change of scale from the solution of flow in an isotropic aquifer. It is, therefore,

sufficient to solve only the latter case and to use the dimensionless variables ϕ' , r' , z' , t' , Q' defined in the list of symbols to cover both isotropic and anisotropic cases, if the well is replaced by a sink line.

Although the linearized solution improves the analysis of pumping tests based on the Theis formula, it is still limited by the assumptions enumerated at the beginning of this section and by the requirement of small drawdowns (i.e., small ϵ).

THE APPROXIMATIVE SOLUTIONS¹

The General Equations

The exact equations satisfied by the velocity potential ϕ' in the case of an aquifer of finite depth (Figure 1) are

$$\nabla^2 \phi' = 0 \quad \text{in the flow domain } -1 < z' < \eta' \quad (2)$$

$$\left. \begin{aligned} \frac{\partial \phi_2'}{\partial t'} + \frac{\partial \phi_2'}{\partial z'} &= \phi_1' \frac{\partial}{\partial z'} \left(\frac{\partial \phi_1'}{\partial t'} + \frac{\partial \phi_1'}{\partial z'} \right) - \left(\frac{\partial \phi_1'}{\partial r'} \right)^2 - \left(\frac{\partial \phi_1'}{\partial z'} \right)^2 \\ \eta_2' &= \phi_1' \frac{\partial \phi_1'}{\partial z'} - \phi_2' \end{aligned} \right\} \quad z' = 0 \quad (16)$$

$$\left. \begin{aligned} \frac{\partial \phi_1'}{\partial t'} + \frac{\partial \phi_1'}{\partial z'} + \left(\frac{\partial \phi_1'}{\partial r'} \right)^2 + \left(\frac{\partial \phi_1'}{\partial z'} \right)^2 &= 0 \\ \eta_2' &= 0 \quad t' = 0 \end{aligned} \right\} \quad z' = -1 \quad (17)$$

$$\left. \begin{aligned} \frac{\partial \phi'}{\partial t'} + \frac{\partial \phi'}{\partial z'} + \left(\frac{\partial \phi'}{\partial r'} \right)^2 + \left(\frac{\partial \phi'}{\partial z'} \right)^2 &= 0 \\ \text{on the water table } z' &= -\eta' \end{aligned} \right\} \quad (3)$$

$$\phi' = -\eta' \quad (4)$$

$$\left. \begin{aligned} \frac{\partial \phi'}{\partial z'} &= 0 \\ \text{on the impervious bottom } z' &= -1 \end{aligned} \right\} \quad (5)$$

$$\left. \begin{aligned} \lim_{r' \rightarrow 0} (r' \frac{\partial \phi'}{\partial r'}) &= -Q'/2\pi l' \\ \text{along the well } -l_2' < z' < -l_1' \end{aligned} \right\} \quad (6)$$

$$\eta' = 0 \quad \text{at } t' = 0. \quad (7)$$

All the variables are dimensionless and are defined in Appendix 1.

Using a small parameter expansion

$$\phi' = Q' \phi_1'(r', z', t')$$

¹ The reader interested mainly in the practical aspects may skip this section.

$$+ Q'^2 \phi_2'(r', z', t') + \dots \quad (8)$$

$$\eta' = Q' \eta_1'(r', t') + Q'^2 \eta_2'(r', t') + \dots$$

the following equations are obtained from 2-7 for ϕ_1' and ϕ_2' [Dagan, 1964]

$$\nabla^2 \phi_1' = 0 \quad \text{in the flow domain } -1 < z' < 0 \quad (9)$$

$$\left. \begin{aligned} \frac{\partial \phi_1'}{\partial t'} + \frac{\partial \phi_1'}{\partial z'} &= 0 \\ \phi_1' &= -\eta_1' \end{aligned} \right\} \quad z' = 0 \quad (10)$$

$$\phi_1' = -\eta_1' \quad (11)$$

$$\frac{\partial \phi_1'}{\partial z'} = 0 \quad z' = -1 \quad (12)$$

$$\left. \begin{aligned} \lim_{r' \rightarrow 0} (r' \frac{\partial \phi_1'}{\partial r'}) &= -1/2\pi l' \\ -l_2' < z' < -l_1' \end{aligned} \right\} \quad (13)$$

$$\eta_1' = 0 \quad t' = 0 \quad (14)$$

$$\nabla^2 \phi_2' = 0 \quad -1 < z' < 0 \quad (15)$$

$$\eta_2' = 0 \quad t' = 0 \quad (18)$$

Hence, in the linearized approximations, equation 3 transforms in the linear b.c. (10) and (16), which are transferred from $z' = -\eta'$ to $z' = 0$.

A general method of solving equations of type 9, 10, 15, 16 by Green functions has been presented by Dagan [1966] and will be used frequently in the following sections.

Aquifer of Infinite Depth

Point-Sink. Variable discharge. In this case it is assumed that the aquifer is very deep ($D \rightarrow \infty$) and that the well reduces to a sink (Figure 2).

The problem being considerably simplified, a more general solution may be obtained by assuming that the discharge is variable $Q = Q(t)$.

With $L = l_2$ as the reference length, and considering only the first approximation $\phi' = Q' \phi_1'$, we represent ϕ' as

$$\phi'(r', z', t') = \frac{Q'(t')}{4\pi} \left\{ \frac{1}{[r'^2 + (z' + 1)^2]^{1/2}} - \frac{1}{[r'^2 + (z' - 1)^2]^{1/2}} \right\} + H(r', z', t') \quad (19)$$

The first term in (19) represents the singularity at the sink, whereas H is harmonic and satisfies, according to (9), (10), (14), and (19), the equations

$$\nabla^2 H = 0$$

$$\text{in the domain } z' < 0 \quad (20)$$

$$\frac{\partial H}{\partial t'} + \frac{\partial H}{\partial z'} = \frac{1}{2\pi} \frac{Q'(t')}{(1 + r'^2)^{3/2}} \quad z' = 0 \quad (21)$$

$$H = 0 \quad z' = 0, \quad t' = 0 \quad (22)$$

The solution of (20)–(22) is expressed by the appropriate Green's function [Dagan, 1966, equation 33] as

$$\begin{aligned} H(r', z', t') &= \frac{1}{2\pi} \int_0^{t'} \int_0^\infty \int_0^\infty \frac{Q'(t' - \tau)}{(1 + \rho_0^2)^{3/2}} e^{\lambda(z' - \tau)} \\ &\quad \cdot J_0(\lambda r') J_0(\lambda \rho_0) \lambda \rho_0 \, d\lambda \, d\rho_0 \, d\tau \end{aligned} \quad (23)$$

After integrating twice in (23) we obtain

$$\begin{aligned} H(r', z', t') &= \frac{1}{2\pi} \int_0^{t'} \frac{(1 + \tau - z') Q'(t' - \tau)}{[(1 + \tau - z')^2 + r'^2]^{3/2}} d\tau = \frac{1}{2\pi} \left\{ \frac{Q'(t')}{[(1 - z')^2 + r'^2]^{1/2}} \right. \\ &\quad \left. - \frac{Q'(0)}{[(t' + 1 - z')^2 + r'^2]^{1/2}} - \int_0^{t'} \frac{dQ'}{dt'} \frac{d\tau}{[(t' - \tau + 1 - z')^2 + r'^2]^{1/2}} \right\} \end{aligned} \quad (24)$$

Hence, ϕ' is expressed, according to (19) and (24), by

$$\begin{aligned} \phi' &= \frac{Q'(t')}{4\pi} \left\{ \frac{1}{[r'^2 + (1 + z')^2]^{1/2}} + \frac{1}{[r'^2 + (1 - z')^2]^{1/2}} \right\} \\ &\quad - \frac{Q'(0)}{2\pi [(t' + 1 - z')^2 + r'^2]^{1/2}} - \frac{1}{2\pi} \int_0^{t'} \frac{dQ'}{dt'} \frac{d\tau}{[(t' - \tau + 1 - z')^2 + r'^2]^{1/2}} \end{aligned} \quad (25)$$

Point sink. Constant discharge. When it is assumed that Q' is constant, (25) and (11) yield for the first approximation ϕ_1' and η_1'

$$\begin{aligned} \phi_1' &= \frac{1}{4\pi} \left\{ \frac{1}{[r'^2 + (1 + z')^2]^{1/2}} \right. \\ &\quad \left. + \frac{1}{[r'^2 + (1 - z')^2]^{1/2}} \right\} \\ &\quad - \frac{1}{2\pi} \frac{1}{[(t' + 1 - z')^2 + r'^2]^{1/2}} \end{aligned} \quad (26)$$

$$\begin{aligned} \eta_1' &= -\frac{1}{2\pi} \frac{1}{(1 + r'^2)^{1/2}} \\ &\quad - \frac{1}{2\pi} \frac{1}{[(1 + t')^2 + r'^2]^{1/2}} \end{aligned} \quad (27)$$

a result similar to that of Boulton [1954].

Since solution 26 is very simple, the second approximation ϕ_2' may be found easily. For this purpose it is necessary to compute the rhs of equation 16, which is noted as $\xi(r', t')$. After carrying out algebraic manipulations, equations 16, 17, and 18 become

$$\begin{aligned} \frac{\partial \phi_2'}{\partial z'} + \frac{\partial \phi_2'}{\partial t'} &= \xi(r', t') \\ &= \frac{1}{4\pi^2} \left\{ \frac{2(1 - r'^2)}{(1 + r'^2)^3} - \frac{1}{[(1 + t')^2 + r'^2]^2} \right. \\ &\quad + \frac{1}{(1 + r'^2)^{3/2}} \frac{1}{[(1 + t')^2 + r'^2]^{1/2}} \\ &\quad - \frac{3}{(1 + r'^2)^{5/2}} \frac{1}{[(1 + t')^2 + r'^2]^{1/2}} \\ &\quad \left. + \frac{2r'^2}{(1 + r'^2)^{3/2} [(1 + t')^2 + r'^2]^{3/2}} \right\} \end{aligned} \quad (28)$$

$$\begin{aligned} \eta_2'(r', t') &= -\frac{1}{4\pi^2} \frac{1 + t'}{[(1 + t')^2 + r'^2]^{3/2}} \\ &\quad \cdot \left[\frac{1}{(1 + r'^2)^{1/2}} - \frac{1}{[(1 + t')^2 + r'^2]^{1/2}} \right] \\ &\quad - \phi_2'(0, r', t') \end{aligned} \quad (29)$$

$$\phi_2'(0, r', 0) = 0 \quad (30)$$

The solution of ϕ_2' , by the adequate Green's

function, is

$$\begin{aligned} \phi_2'(r', z', t') \\ = \int_0^{t'} \int_0^\infty \int_0^\infty e^{\lambda(z'-\tau)} \xi(\rho_0, t' - \tau) \\ \cdot J_0(\lambda r') J_0(\lambda \rho_0) \lambda \rho_0 d\lambda d\rho_0 d\tau \end{aligned} \quad (31)$$

The integration with respect to λ may be carried out for $r' = 0$

$$\begin{aligned} \phi_2'(0, z', t') \\ = \int_0^{t'} \int_0^\infty \frac{\xi(\rho_0, t' - \tau)(z' - \tau) \rho_0 d\rho_0 d\tau}{[(z' - \tau)^2 + \rho_0^2]^{3/2}} \end{aligned} \quad (32)$$

Using (32), the free-surface drop at $r' = 0$ was computed numerically by the aid of the Elliot 803 Technion computer.

Both first and second approximations $\eta_1(0, t')$ and $\eta_2(0, t')$ are represented in Figure 3.

At $t \rightarrow \infty$ a steady state is reached, and $\eta_2'(0, \infty)$ may be computed analytically by the integration of (32), which yields

$$\begin{aligned} \phi_2'(0, 0, \infty) \\ = \int_0^\infty \int_0^\infty \xi(\rho_0, \infty) J_0(\lambda \rho_0) \rho_0 \lambda d\lambda d\rho_0 = 1/16\pi \end{aligned} \quad (33)$$

Using (27) and (29), we obtain

$$\begin{aligned} \eta_1'(0, \infty) &= -1/2\pi \\ \eta_2'(0, \infty) &= -1/16\pi \end{aligned} \quad (34)$$

Therefore, the water-table drop expressed in a dimensional form is, by (8) and (34)

$$\eta(0, \infty) = -(Q/2\pi K_h l_3^2) - (Q^2/16\pi K_h^2 l_3^4) \quad (35)$$

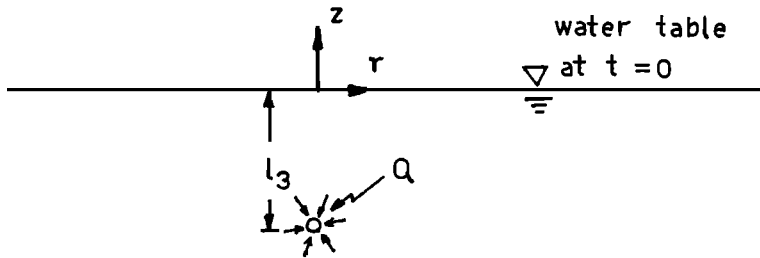


Fig. 2. A sink in a deep aquifer.

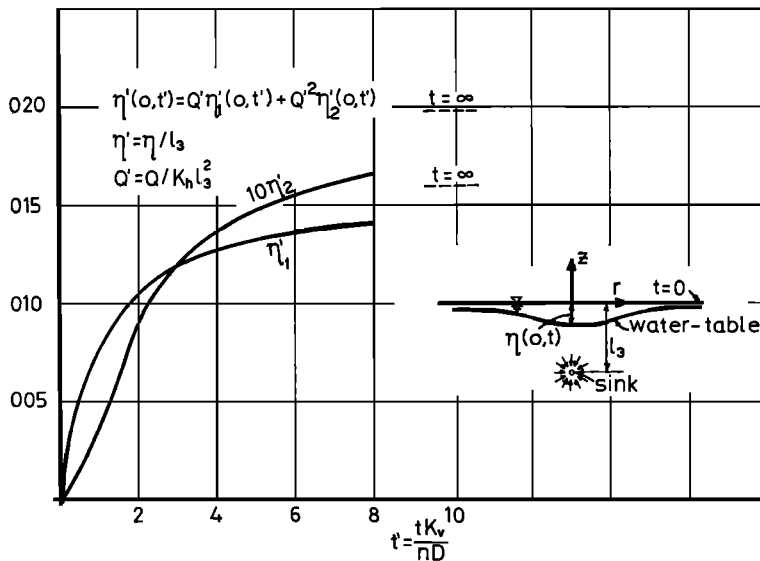


Fig. 3. First and second approximation of the drawdown (deep aquifer).

If it is required, for instance, that the second approximation will not exceed 0.05η , the following upper limit for Q is obtained:

$$Q = 0.422 K_h l_3^2 \quad (36)$$

In most conditions the discharge is smaller than this upper bound.

In the case of an aquifer of finite thickness or of a well of finite length, the computation of the second approximation becomes very tedious. Therefore, only first approximations ϕ_1' will be derived in the following sections.

Well of finite length. Constant discharge. In the case of a well of finite length l in a deep

aquifer ($D/l_2 \rightarrow \infty$) $L = l_2$ is taken as reference length (Figure 1).

The solution obtained by replacing the well by a sink line of uniform strength is approximate, since the condition of constant head on the well perimeter at each instant is not satisfied [Muskat, 1937]. However, the head distribution throughout the aquifer at points not too close to the well is still accurate.

The solution in this case is obtained by the summation of elementary sinks distributed along the well. Taking ϕ_1' from (26) as an elementary solution, and considering the uniform strength of the sink equal to Q'/l' , we obtain

$$\phi_1' = \frac{1}{4\pi l'} \int_{l_1'}^{l_2'} \left\{ \frac{1}{[r'^2 + (z' + \xi')^2]^{1/2}} + \frac{1}{[r'^2 + (z' - \xi')^2]^{1/2}} - \frac{2}{[(t' + \xi' - z')^2 + r'^2]^{1/2}} \right\} d\xi' \quad (37)$$

The integration of (37) yields

$$\begin{aligned} \phi_1' = \frac{1}{4\pi l'} \left[\ln \frac{l_2' + z' + \sqrt{(l_2' + z')^2 + r'^2}}{l_1' + z' + \sqrt{(l_1' + z')^2 + r'^2}} + \ln \frac{l_2' - z' + \sqrt{(l_2' - z')^2 + r'^2}}{l_1' - z' + \sqrt{(l_1' - z')^2 + r'^2}} \right. \\ \left. - 2 \ln \frac{l_2' + t' - z' + \sqrt{(l_2' + t' - z')^2 + r'^2}}{l_1' + t' - z' + \sqrt{(l_1' + t' - z')^2 + r'^2}} \right] \quad (38) \end{aligned}$$

a result similar to that of Boulton [1954].

If the solution is approximated only by the first-order term $\phi' = Q'\phi_1'$, the water table shape is described by

$$\begin{aligned} \eta' = -\phi'(r', 0, t') \\ = -\frac{Q'}{2\pi l'} \left[\ln \frac{l_2' + \sqrt{l_2'^2 + r'^2}}{l_1' + \sqrt{l_1'^2 + r'^2}} - \ln \frac{l_2' + t' + \sqrt{(l_2' + t')^2 + r'^2}}{l_1' + t' + \sqrt{(l_1' + t')^2 + r'^2}} \right] \quad (39) \end{aligned}$$

Aquifer of Finite Thickness

Sink of constant discharge. Only the first-order solution is considered, and the aquifer depth is taken as reference length, i.e., $L = D$.

The solution for the potential $\phi' = Q'\phi_1'$ is represented by the Green's function already derived by Dagan [1966, equation 31]

$$\begin{aligned} \phi'(r', z', t') = \frac{Q'}{4\pi} G(z', r', -l_3', t') = \frac{Q'}{4\pi} \left\{ \frac{1}{[r'^2 + (z' + l_3')^2]^{1/2}} + \frac{1}{[r'^2 + (z' - l_3')^2]^{1/2}} \right. \\ \left. - 2 \int_0^\infty \frac{\cosh \lambda(1 - l_3') \cosh \lambda(1 + z') e^{-\lambda t' \tanh \lambda}}{\sinh \lambda \cosh \lambda} J_0(\lambda r') d\lambda \right. \\ \left. + 2 \int_0^\infty \frac{e^{-\lambda} \cosh \lambda l_3' \cosh \lambda z'}{\sinh \lambda} J_0(\lambda r') d\lambda \right\} \quad (40) \end{aligned}$$

Well of finite length. Constant discharge. Again it is assumed that the well may be represented as a sink line of uniform strength. The reference length is $L = D$. The solution is obtained from (40) by integration along the well. The elementary uniformly distributed discharge of the sinks is Q'/l . Hence

$$\phi(r', z', t') = \frac{Q'}{4\pi l'} \int_{l_1'}^{l_2'} \left\{ \frac{1}{[r'^2 + (z' + \xi)^2]^{1/2}} + \frac{1}{[r'^2 + (z' - \xi)^2]^{1/2}} \right. \\ \left. - 2 \int_0^\infty \frac{\cosh \lambda(1 - \xi) \cosh \lambda(1 + z') e^{-\lambda t' \tanh \lambda}}{\sinh \lambda \cosh \lambda} J_0(\lambda r') d\lambda \right. \\ \left. + 2 \int_0^\infty \frac{e^{-\lambda} \cosh \lambda \xi \cosh \lambda z'}{\sinh \lambda} J_0(\lambda r') d\lambda \right\} d\xi \quad (41)$$

The integration of (40) with respect to ξ and the replacement of $l_1' = l_3' - l'/2$ and $l_2' = l_3' + l'/2$ yields

$$\phi' = \frac{Q'}{\pi l'} \left\{ \frac{1}{4} \ln \frac{l_3' + \frac{l'}{2} + z' + \left[\left(l_3' + \frac{l'}{2} + z' \right)^2 + r'^2 \right]^{1/2}}{l_3' - \frac{l'}{2} + z' + \left[\left(l_3' - \frac{l'}{2} + z' \right)^2 + r'^2 \right]^{1/2}} \right. \\ \cdot \frac{l_3' + \frac{l'}{2} - z' + \left[\left(l_3' + \frac{l'}{2} - z' \right)^2 + r'^2 \right]^{1/2}}{l_3' - \frac{l'}{2} - z' + \left[\left(l_3' - \frac{l'}{2} - z' \right)^2 + r'^2 \right]^{1/2}} \\ \left. - \int_0^\infty \frac{\cosh \lambda(1 + z') \sinh \lambda \frac{l'}{2} \cosh \lambda(1 - l_3')}{\lambda \sinh \lambda \cosh \lambda} e^{-\lambda t' \tanh \lambda} J_0(\lambda r') d\lambda \right. \\ \left. + \int_0^\infty \frac{e^{-\lambda} \sinh \lambda \frac{l'}{2} \cosh \lambda l_3' \cosh \lambda z'}{\lambda \sinh \lambda} J_0(\lambda r') d\lambda \right\} \quad (42)$$

The solution 42 has been used to compute numerically $\phi'/Q' = sK_b D/Q$ as a function of the variables r' , z' , t' and the parameters l_3' , l' .

DISCUSSION OF RESULTS AND OUTLINE OF THE MATCHING METHOD

Two of the five cases solved in the previous section are of particular interest.

The solution of flow towards a point sink in an aquifer of infinite depth (equation 35) is of theoretical interest, since it provides two terms in the expansion of ϕ' in a Q' power series. The computation of the water table drawdown at $r = 0$ (Figure 3) shows that the expansion seems to be convergent if Q' is sufficiently small. In this case the first-order term may represent the solution quite accurately.

From a practical point of view, the solution of flow towards a partially penetrating well in an aquifer of finite depth (equation 42) is particularly important. It relates the dimensionless drawdown $s' = \phi'/Q' = sK_b D/Q$ to the

variables $z' = z/D$, $r' = (K_s/K_b)^{1/2} r/D$, $t' = tK_b/nD$ and the parameters $l' = l/D$ and $l_3' = l_3/D$ (Figure 1). This solution is valid in the vicinity of as well as at large distance from the pumping well. However, it cannot be used on the well perimeter or at a distance of a few well diameters, because the well has been replaced by a sink line. The solution is accurate only when Q' or the drawdowns are small, and it assumes that the discharge and the effective porosity are constant. It is also assumed that the observation well has a relatively short filter and the drawdown is measured at its center (otherwise one has to average the head along the filter).

Values of s' as expressed by (42) have been computed numerically. The integration, according to Simpson rule, has been carried out by dividing each interval between two zeros of the Bessel function into 30 subintervals. The numerical procedure has been used in the analysis of the examples in the section below. Ex-

tensive tables relating s' to different values of the parameters and the computer program are presented in the report by Dagan [1967a].

The comparison of the s' values as resulting from equation 42 and s' as computed from the Theis formula shows that for $r' > 1.20$ and $t' > 6$ the difference between the two s' becomes smaller than 4%, even for $z' = -0.05$. This difference is larger in the case of small l' , but even in the case of fully penetrating wells ($l' = 1$) the Theis formula provides s' values different from the solution (equation 42) when $t' < 1$. Therefore, the use of the Theis formula is drastically limited by the requirement $r/D > 1.20 \times (K_a/K_e)^{1/2}$ in the case of anisotropic aquifers.

The solution (equation 42) satisfies linear equations. The superposition principle may therefore be used to determine the drawdown during recovery or to sum the drawdown caused by two wells of different lengths.

The characteristics of the aquifer K_a , K_e , and n may be obtained by matching the measured drawdowns and the theoretical curve (equation 42).

For this purpose the values of l' , l'_e (pumping well), and z' (observation well) must be inserted in equation 42; s' becomes a function of t' and r' for each observation well. A set of curves $s'(t')$ for different constant r' is to be drawn on logarithmic paper. Since most anisotropic aquifers have $K_e/K_a < 1$, r' has to be smaller than r/D .

The measured drawdowns must be represented on similar logarithmic paper. By matching them with one of the curves of the set, three values are obtained: r' (the best fitting curve) and s' , t' (the matching point). Since r , D , Q , and t are known, the values of K_a , K_e , and t may be easily found. The procedure is exemplified in the following section.

EXAMPLES OF ANALYSIS OF PUMPING TESTS

The data concerning pumping tests carried out in three phreatic aquifers in the United States are analyzed to exemplify the use of the method presented in the previous chapters.

Wenzel [1942]. The data collected by Wenzel from the pumping test at Grand Island, Nebraska, are first discussed. Since the measurements in 83 observation wells are presented in detail, the data are particularly suitable for the analysis.

The general data of the aquifer and the test are: $D = 100$ feet, $Q = 2940$ mc/day, $l_e = 19.8$ feet, $l = 39.5$ feet. Therefore, $l' = l/D = 0.395$, $l'_e = l_e/D = 0.198$.

In a first stage 14 observation wells were selected to compute the aquifer characteristics. They are identified in Table 1A, columns 1, 2, 3. The distance from the pumping well (r) varies between 184.9 feet (well 16B) to 516.5 feet (well 63 SW). Using the values of l'_e and l' of the aquifer and z' of the observation wells in equation 42, curves relating $\phi'/Q' = s' = sK_a D/Q$ to $t' = tK_e/nD$ at different values of $r' = (r/D)(K_e/K_a)^{1/2}$ have been computed numerically and drawn on logarithmic paper (Figure 4).

The drawdowns s as a function of t have been drawn on similar logarithmic paper, and the measured points and theoretical curves have been matched (Figure 4) according to the method presented in the preceding section.

Drawdowns of less than 0.05 feet have been discarded, since the measurements do not seem reliable when $s < 0.05$ feet. The values of r' , s' , t' , s , t at the matching points are presented in Table 1A (columns 4, 5, 6, 7, 8) and the resulting K_e/K_a , K_a , K_e and n in columns 9, 10, 11, 12. The results show an average anisotropy of $K_e/K_a = 0.13$ and $K_a = 140$ m/day. Wenzel [1942, p. 142] has determined K by Theis non-equilibrium formula and has found $K \equiv 41$ m/day. This K should be equal to the above K_a , but the discrepancy is very large. A further investigation of Wenzel's computations has shown that he has used data from observation wells at $r = 50 \div 800$ feet, i.e., at $r' = (K_e/K_a)^{1/2} r/D = 0.18 \div 2.88$ [Wenzel, 1942, p. 124] for matching with the Theis formula. Hence most of Wenzel's points are at a distance from the pumping well for which the Theis formula is not valid; this error becomes recognizable when the anisotropy is determined.

We have also matched observations with the Theis type curve, but only for wells at $r = 604 \div 716.5$ feet (six wells, Table 1B). Each well has been matched separately. The type curve and the measured points are given in Figure 5.

The results are collected in Table 1B, and the K_a values (column 18) are in agreement with the results obtained in column 11.

The average values obtained from the 18 ob-

servation wells of Table 1, Figures 4 and 5, and the average root mean square of the deviations are

$$\begin{aligned} K_h &= 141 \text{ m/day} \quad (\pm 4.2\%) \\ K_v &= 18.2 \text{ m/day} \quad (\pm 9\%) \\ n &= 0.141 \quad (\pm 11\%) \end{aligned}$$

Whereas the values of K_h and K_v are erratically distributed, a systematic increase of n with the distance became apparent (Figure 6).

The values at large distance from the pumping well are in agreement with those obtained by Wenzel. It is possible that the average value

$n = 0.141$ based on Table 1 is somewhat lower than the real average, since most of the points are at $r < 500$ feet.

Stallman [1963] has tried to determine the characteristics of the same aquifer by the aid of an electric analog. The analog, in fact, provides the first order linearized solution at $t = 0$. However, the measurements at small t are not reliable. This is probably the cause of the difficulties encountered by Stallman in the attempt to interpret the data in this way.

Weeks (1964). Weeks has tried to determine the anisotropy and the aquifer characteristics at

TABLE 1. Wenzel [1942] Pumping Test at Long Island, Nebraska
A. Observation Wells at $r < 600$ feet

Well No.	r , ft	$-z$, ft	Matching Point (Fig. 4)					Results			
			Theoretical			Measured		K_v/K_h	K_v , m/day	K_h , m/day	n
			$r/D(K_v/K_h)^{1/2}$	$sK_h D/Q$	tK_v/nD	s , ft	t , min				
1	2	3	4	5	6	7	8	9	10	11	12
7A	429.3	5.08	1.7	0.1	4.3	0.20	600	0.157	24.8	158	0.0786
16B	184.9	4.82	0.5	0.1	0.35	0.34	160	0.073	6.77	92.6	0.0703
17B	254.7	5.04	0.7	0.1	0.73	0.31	320	0.076	7.75	102	0.0766
19B	424.6	5.27	1.7	0.1	4.3	0.20	800	0.161	24.9	158	0.1035
20B	499.7	4.82	1.7	0.1	4.3	0.17	1450	0.116	21.6	186	0.1660
27W	270.0	5.38	0.9	0.1	1.2	0.29	350	0.111	12.1	109	0.0804
48C	285.0	5.00	1.3	0.1	2.5	0.29	400	0.207	22.5	109	0.0820
61SW	316.6	5.10	1.3	0.1	2.5	0.27	540	0.169	19.7	117	0.0968
63SW	516.6	5.10	1.7	0.1	4.3	0.19	1800	0.108	17.9	166	0.1710
81N	445.8	5.25	1.7	0.1	4.3	0.20	1000	0.145	22.9	138	0.1210
8A	478.9	6.15	1.8	0.1	4.5	0.20	800	0.141	22.3	158	0.0907
18B	375.3	5.88	1.4	0.1	2.8	0.24	720	0.138	18.2	132	0.1060
76S	382.7	5.99	1.4	0.1	2.8	0.19	440	0.134	22.3	166	0.0800
80N	342.0	6.13	1.0	0.1	1.45	0.25	500	0.0845	10.7	126	0.0840

B. Observation Wells at $r > 600$ feet

Well No.	r , ft	$-z$, ft	Matching Point (Fig. 5)				Results	
			Theoretical		Measured		K_h , m/day	n
			$tK_h D/nr^2$	$sK_h D/Q$	s , ft	t , min		
1	2	3	13	14	15	16	17	18
9A	604.0	5.58	0.600	0.055	0.11	1000	158.0	0.164
10A	754.6	5.82	0.500	0.046	0.10	2000	145.5	0.133
21B	649.7	4.70	0.390	0.033	0.10	1750	104.0	0.249
30W	804.5	6.25	0.185	0.010	0.019	770	166.5	0.242
40D	722.7	7.02	0.500	0.046	0.100	2000	145.5	0.252
65W	716.5	4.45	0.580	0.050	0.100	2200	158.0	0.174

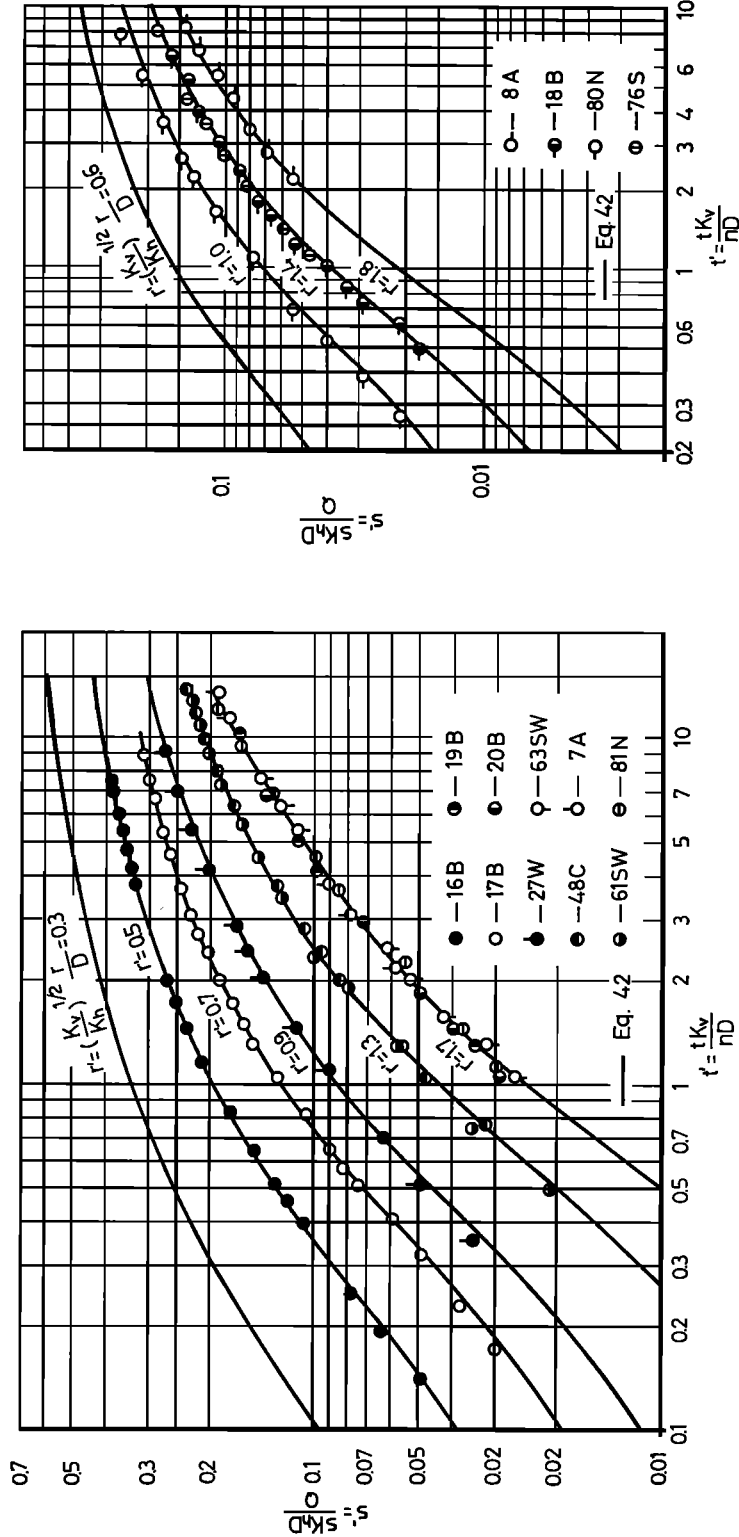


Fig. 4. The analysis of Wenzel [1942] data at Long Island, Nebraska.

Portage County, Wisconsin, by using the *Hantush* [1964] solution for partially penetrating wells in confined elastic aquifers. The use of this solution in the case of phreatic aquifers is questionable, since in elastic aquifers water is released in each point, whereas in phreatic aquifers it is released only at the water table. The matching of the measured data with both *Hantush* and *Theis* formulas [Weeks, 1964, Figure 3] is not accurate, and the n values are contradictory.

Unfortunately, the data concerning the aquifer and the observations are not given in tables, and to analyze the pumping test we have been forced to use the figures only. It was estimated that $D = 80$ feet, $Q = 5780$ mc/day, $l_s = 71$ feet, and $l = 17.6$ feet.

The data obtained in the observation well 2S ($r = 100$ feet, $z = -23$ feet) [Weeks, Figure 7] have been used, tentatively, matched with the theoretical curve (equation 42). The result is shown in Figure 7.

The matching curve has $r' = (K_s/K_a)^{1/2} r/D = 0.9$; at the matching point $sK_a D/Q = 0.1$, $s = 1$ foot, $tK_a/nD = 0.95$, $t = 0.21$ day. Therefore, $K_a \approx 78$ m/day, $K_s \approx 40$ m/day, $n = 0.36$, $K_s/K_a = 0.52$. These results are in discrepancy with *Week's* results ($K_s/K_a = 0.05$, $K_a = 61.5$ m/day, $n = 0.12 \div 0.16$).

Mansur and Dietrich [1965]. *Mansur and Dietrich* [1965] have tried to determine the permeabilities ratio of an anisotropic aquifer at the Arkansas River by two methods of interpretation of the pumping test: by an electrolytic model and by *Muskat* solution for a partially penetrating well in a confined aquifer. Both methods are based on a steady state.

Unfortunately, the data of the observation wells are given only graphically [Mansur and Dietrich, Figure 9]. The data from the well D_s have been used, tentatively, for matching. The general data are: $D = 88$ feet, $Q = 8190$ mc/day, $l = 88$ feet (fully penetrating well), $l_s = 44$ feet, $r = 40$ feet, $z = -24$ feet.

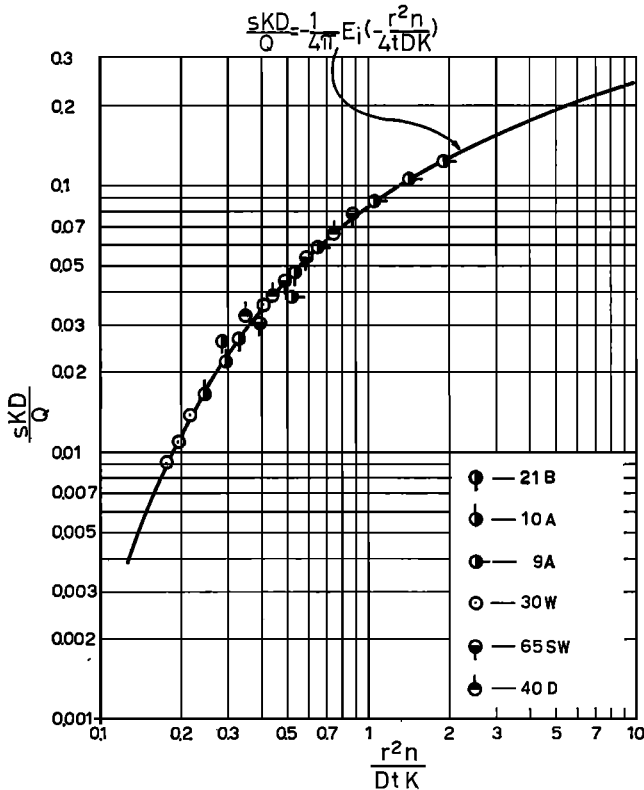


Fig. 5. The analysis of *Wenzel* [1942] data by the *Theis* formula.

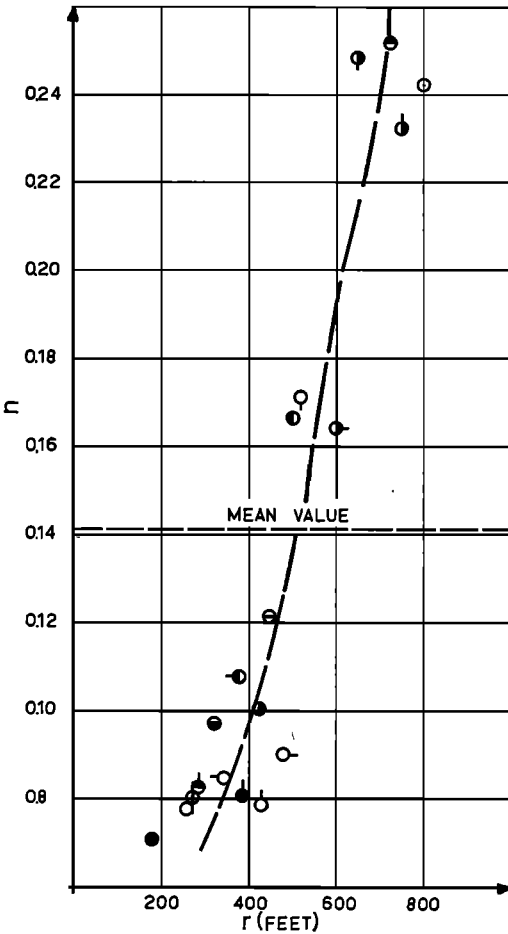


Fig. 6. The variation of n with r as obtained from Wenzel [1942] data.

The theoretical curves and experimental points are given in Figure 8. The matching curve has $r' = (K_v/K_h)^{1/2} r/D = 0.3$; at the matching point $sK_h D/Q = 0.19$, $s = 4.55$ feet, $tK_h/nD = 0.23$, $t = 20$ h. Therefore, $K_h \approx 41$ m/day, $K_v = 19.5$ m/day, $n = 0.264$, $K_h/K_v = 2.13$. Both K_h and K_h/K_v values compare favorably with the results obtained by Mansur and Dietrich ($K_h = 56$ m/day, $K_h/K_v \approx 2$).

CONCLUSIONS

The first-order linearized solution of phreatic flow to partially penetrating wells provides a method of evaluating the permeability and effective porosity of the unconfined aquifer. This method is an improvement of the actual techniques, since it takes into consideration the vertical components of the flow velocity. However, the usual assumptions of constant effective porosity and small drawdowns are still adopted.

Acknowledgment. This research work has been carried out at the Hydraulic Laboratory of the Technion Israel Institute of Technology and has been sponsored by Mekoroth Water Company, Ltd. and Tahal Water Planning for Israel, Ltd.

REFERENCES

- Bear, J., and G. Dagan, The relationship between solutions of flow problems in isotropic and anisotropic soils, *J. Hydrol.*, 9, 23-31, 1965.
 Boulton, N. S., The drawdown of the water table under nonsteady conditions near a pumped well in an unconfined formation, *Inst. Civil Eng. Proc.*, pt. 3, 564-579, 1954.

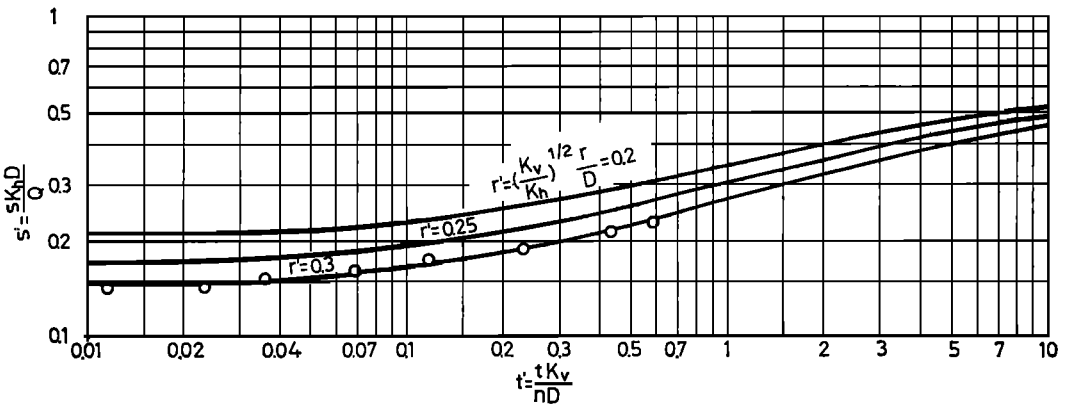


Fig. 7. The analysis of Weeks [1964] data at Portage County, Wisconsin.

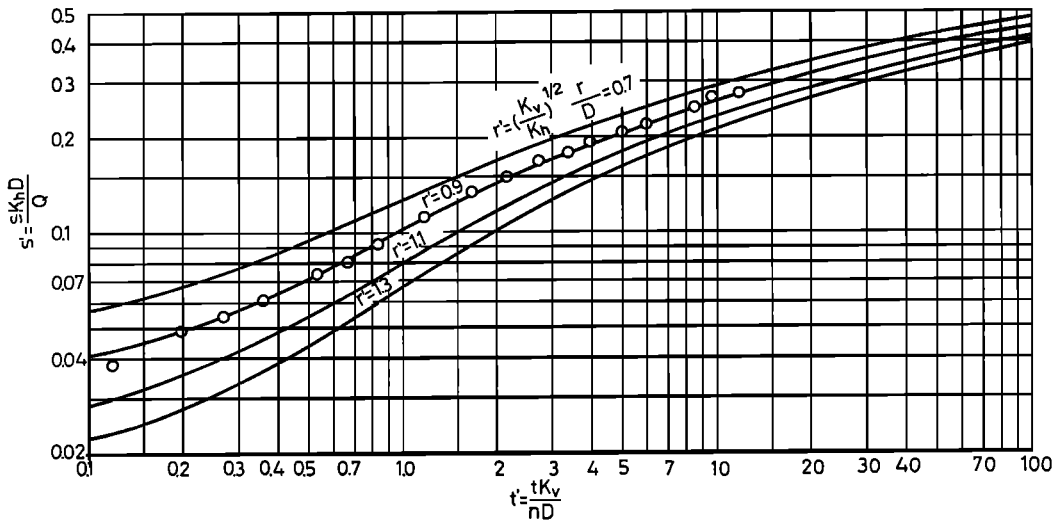


Fig. 8. The analysis of Dietrich and Mansur [1965] data at Arkansas River.

- Dagan, G., Second order linearized theory of free surface flow in porous media, *La Houille Blanche*, Grenoble, France, 8, 901-910, 1964.
- Dagan, G., The solution of the linearized equations of free-surface flow in porous media, *J. Mecanique*, 5(2), 207-215, 1966.
- Dagan, G., A method of determining the permeability and effective porosity of unconfined anisotropic aquifers, *Hydraul. Lab., Technion-Israel Inst. Technol. P.N. 1/1967*, p. 49, January 1967a.
- Dagan, G., Solutions of free-surface flow in porous media, *Proc. Am. Soc. Civil Engrs., Div. Eng. Mech.*, Aug., 1967b.
- Hantush, S. M., Hydraulics of wells, in *Advances in Hydrosience*, edited by Ven T. Chow, Academic Press, New York and London, 1, 282-430, 1964.
- Kirkham, D., Seepage of steady rainfall through soils into drains, *Trans. Am. Geophys. Union*, 39(5), 892-908, 1958.
- Mansur, C. J., and R. J. Dietrich, Pumping test to determine permeability ratio, *Proc. Soil Mechanics and Foundations Div., SM4*, 151-183, 1965.
- Muskat, M., *The Flow of Homogeneous Fluids through Porous Media*, McGraw-Hill Book Company, New York, 1937.
- Polubarinova-Kochina, P. Ya., *Theory of Ground Water Movement* (translated from Russian by J. M. R. De Wiest), Princeton University Press, 613 pp., 1962.
- Stallman, R. W., Electric analog of three-dimensional flow to wells and its application to unconfined aquifers, *U. S. Geol. Surv. Water-Supply Paper 1536-H*, 205-242, 1963.
- Stallman, R. W., Effects of water table conditions on water level changes near pumping wells, *Water Resources Res.*, 1(2), 295-311, 1965.
- Theis, C. V., Relation between the lowering of the piezometric surface and the rate and duration of discharge of a well using groundwater storage, *Trans. Am. Geophys. Union*, 14, 519-524, 1935.
- Weeks, E. P., Field methods for determining vertical permeability and aquifer anisotropy, *U. S. Geol. Surv., Prof. Paper 501-D*, D 193-D 198, 1964.
- Wenzel, L. K., Methods for determining permeability of water bearing materials, *U. S. Dept. Interior, Geol. Surv. Water-Supply Paper 837*, 192 pp., 1942.

(Manuscript originally received March 7, 1967; revised June 12, 1967.)